

Maxim (Max) Khanov

Senior undergraduate computer science student interested in deep learning, particularly **LLMs** and **computer vision**. Seven publications in various conferences including **ICLR**, **ICML**, **ICMLA**. **Google Scholar Citations: 156**. **h-index: 5**.

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Education

University of Wisconsin-Madison (2022 - 2025)

BS, Computer Science (GPA: 3.987/4.0)

- **Completed Coursework:** Intro-Theory of Probability, Theory and Algorithms for Data Science, Deep Learning for Computer Vision, Intro to Learning Theory, Machine Learning (graduate level), Advanced Topics in Reinforcement Learning (graduate level)

Research Experience

Research under Dr. Sharon Yixuan Li (2023 - 2025)

- Worked primarily in the area of **LLMs**, with an interest in **AI Alignment** and **efficiency**
- Co-authored **two** papers (both co-first authorship), which were accepted into **ICLR** [2], and **ICML** [1]
- Participated in weekly meetings and the internal paper review process of **three** papers which have been accepted to **ICLR** and **ICML**

Research under Dr. Esra Akbas

- Worked in areas such as **graph neural networks**, **computer vision**, and **NLP**, designing novel architectures to improve performance metrics
- Co-authored five papers, which were accepted into venues such as **PAKDD**, and **ICMLA** [3-7]
- Participated in weekly meetings which included paper presentations

Teaching Experience

Random Signal Analysis In Class Peer Mentor under Dr. Eduardo Arvelo (2024)

- Helped in class and assisted teaching students probability theory, signal analysis, and statistical methods.
- Covered topics: introductory probability theory, parameter estimation, random processes, basic statistical learning theory (empirical risk minimization)

Publications

(Asterisk: [Co-] First Author; Underline: Advisor)

[1] How Contaminated Is Your Benchmark? Quantifying Dataset Leakage in Large Language Models with Kernel Divergence (ICML), 2025

Hyeon Kyu Choi*, **Maxim Khanov***, Hongxin Wei, Yixuan Li

- We introduced **KDS** which takes a distributional perspective for quantifying dataset contamination in LLMs

[2] ARGS: Alignment as Reward-Guided Search (ICLR), 2024

Maxim Khanov*, Jirayu Burapachee*, Yixuan Li

- Introduced **ARGS** which reframes the LLM alignment problem as an inference time reward guided search
- Paper achieved **top 5%** reviewer score **across >7000** submissions
- As of Jan 4th, 2025 has **121 Citations**
- When checking implementation, discovered and reported a bug in Hugging Face Transformers

[3] MPool: Motif-Based Graph Pooling (PAKDD), 2023

Muhammad Ifta Khairul Islam, **Max Khanov**, Esra Akbas

- Introduced **MPool**, a graph pooling method which uses motifs to better capture higher order structure in graphs
- Found and corrected a bug in method implementation and implemented/ran experiments

[4] DECADE-deep learning based content-hiding application detection system for Android (IEEE Big Data 2021)

Mingming Peng, **Max Khanov**, Saikeerthi Reddy Madireddy, Hongmei Chi, Esra Akbas, Gokila Dorai

[5] The Need for Biometric Anti-spoofing Policies: The Case of Etsy (ICDF2C 2022)

M Jozani, G Zanella, **M Khanov**, G Dorai, E Akbas

[6] Effects of covid-19 on individuals in opioid addiction recovery (ICMLA 2021)

KM Saifuddin, E Akbas, **M Khanov**, J Beaman

[7] Approximating Dense ReLU Networks Using Linear Regression (KDD-UC 2022)

M Khanov*, E Akbas*

Honors / Awards

Barry M. Goldwater Scholarship, 2024

- National scholarship which recognizes undergraduate research in STEM.

Invited ICLR Reviewer, 2024 *

ISEF Regeneron Finalist (National Research Competition), 2021

** - Rejected due to scheduling constraints*

Oral Presentations

- **ICLR 2024** (*virtual*)
- **KDD-UC 2022**
- Oklahoma Academy of Science Technical Meeting 2021/2020

Skills

- **Languages:** C, C#, Java, Python
- **Machine Learning Frameworks:** Pyro, PyTorch, Hugging Face Transformers, Keras, Jax, Tensorflow, Sklearn, XGBoost
- **Misc:** Streamlit.io, Flask, Numpy, Scipy, Open CV, Uiua